

Leaning Over to Reach Far-Away Objects

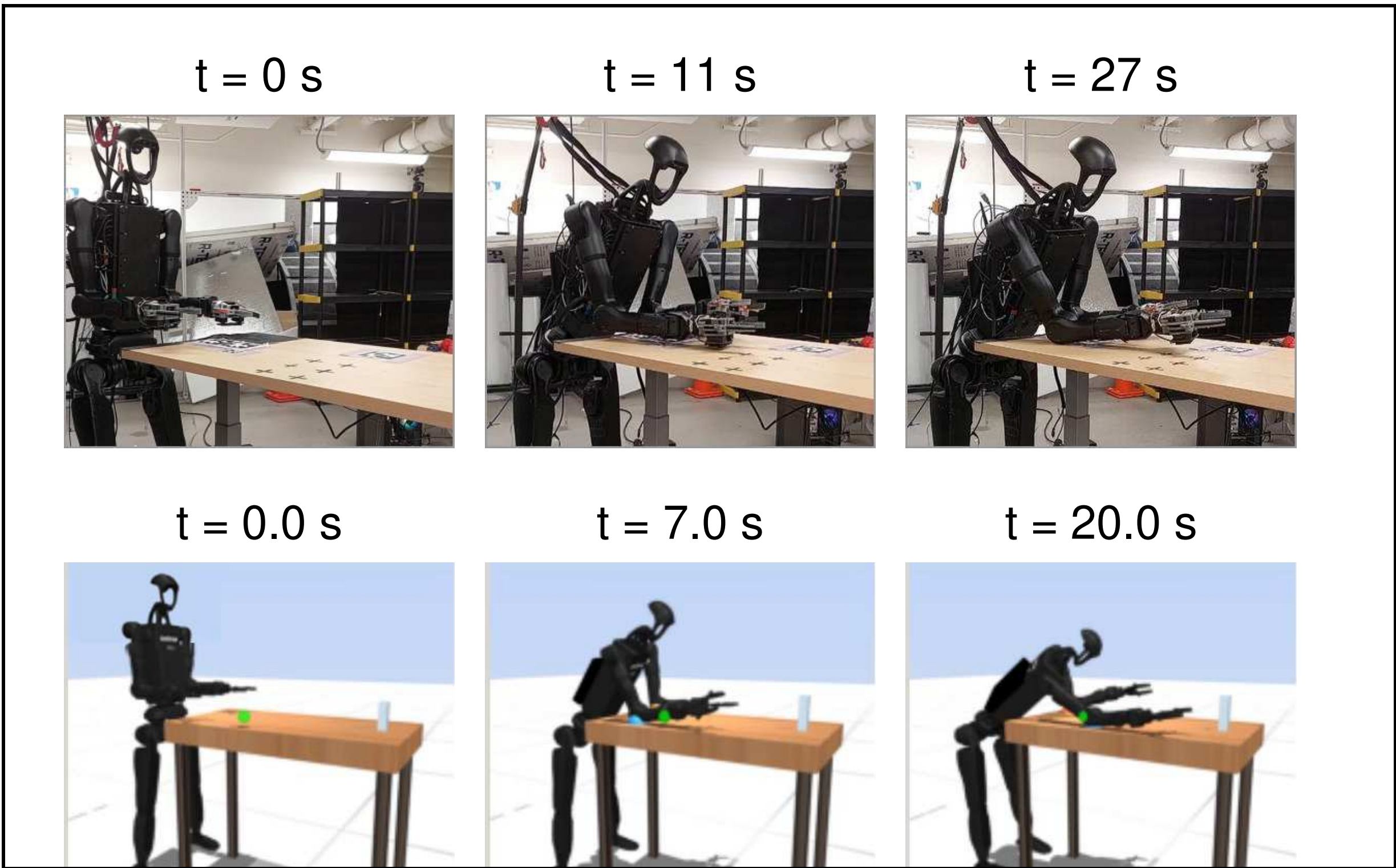
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h12leaner.github.io

Supported reaching on a Unitree H1-2



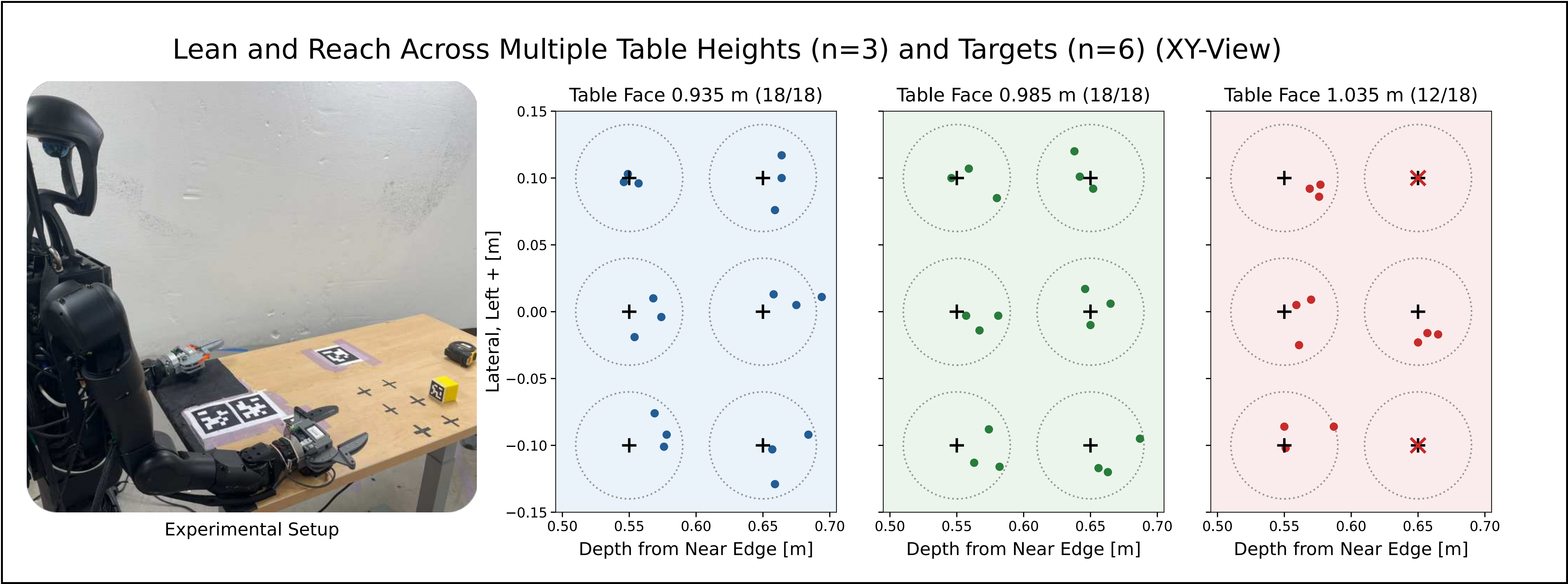
Stand, lean and brace, reach: hardware (top) and MuJoCo (bottom).



The six commanded targets on the 0.985 m table: 82 and 92 cm past the near edge, ± 10 cm laterally.

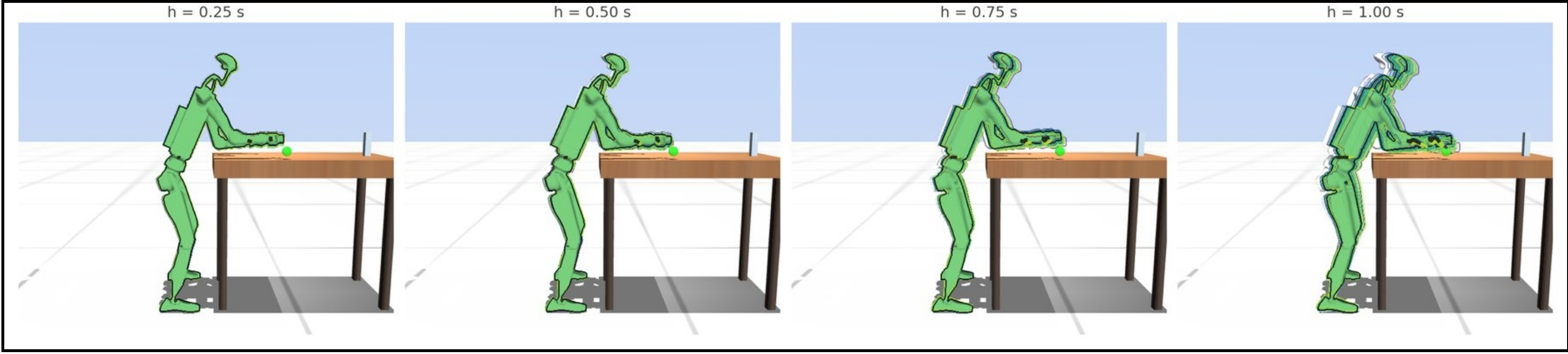
- +50 cm beyond unbraced reach
- 4 min loaded brace sustained
- 89 % reach the target (48/54)
- 81 % full reach-and-recover cycle (44/54)
- 2.7 cm mean tip error, 48 holds
- 34 Hz planning on a laptop CPU

Six targets, three table heights, three trials each, one controller



Measured hold positions (dots) vs. commanded targets (crosses) at 0.935, 0.985 and 1.035 m; circles are 4 cm. 18/18 holds at the two lower heights, 12/18 at the tallest: the two deepest lateral targets collapse during the initial lean.

Sampling-based MPC with off-the-shelf CEM

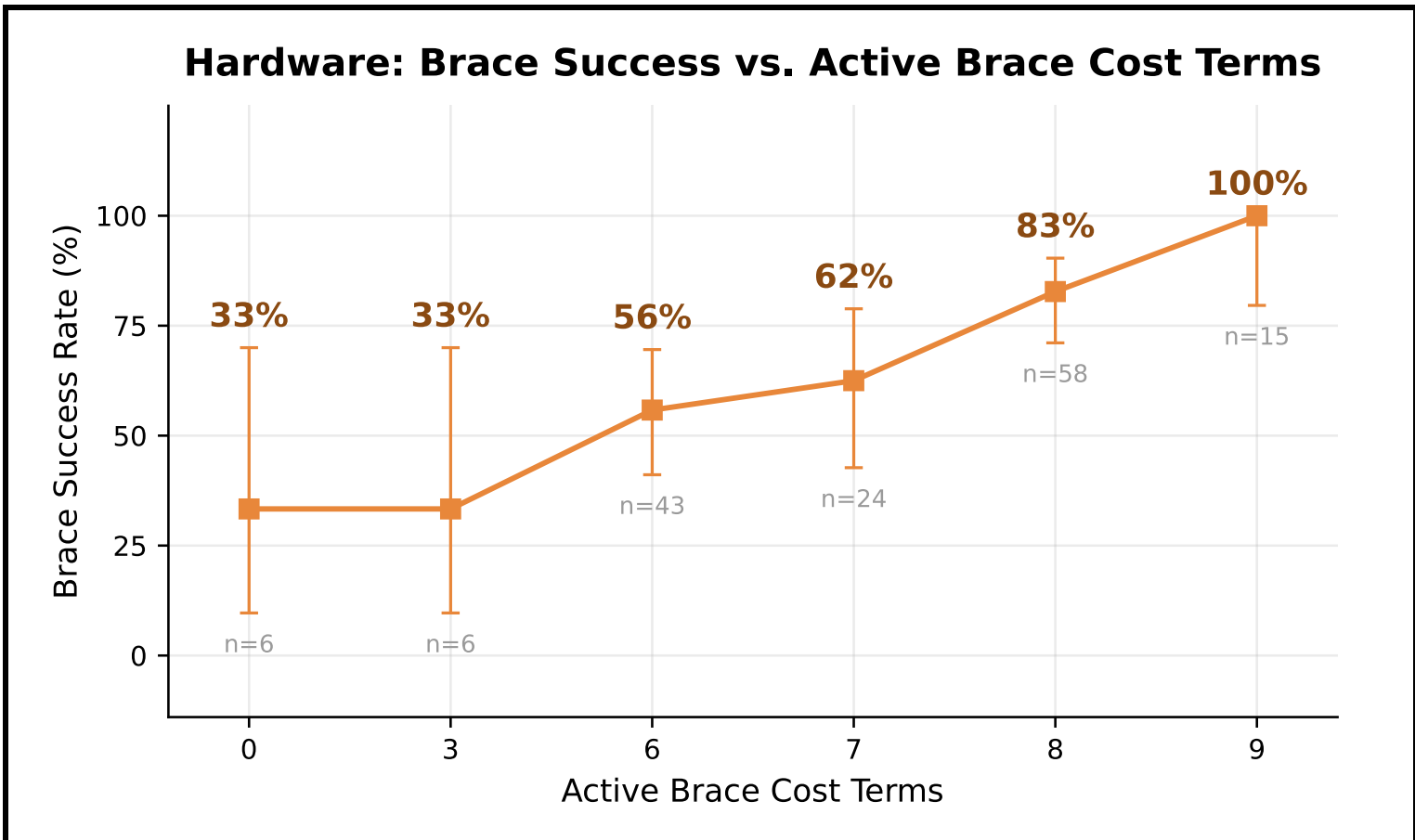


Twenty rollouts from one planning tick (MJPC), coloured by cost. Every 30 ms: sample 20 control splines, roll each out 1 s in MuJoCo, keep the best. Brace terms encourage the forearm to seat on the table; keep-out terms hold every other link away.

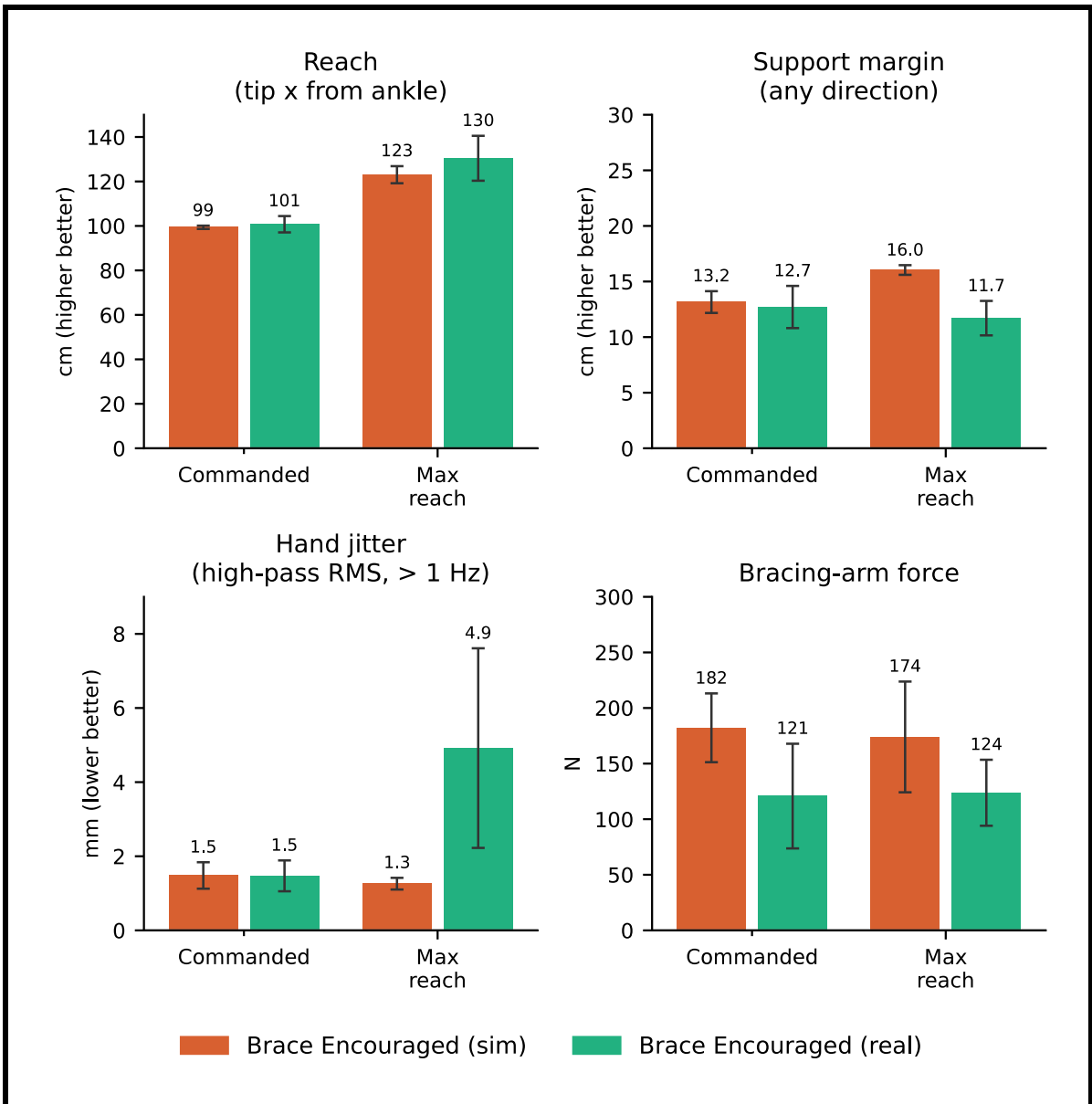
Base estimator	Drift (horiz.)	Height
IMU + leg odometry	+267 mm/min	1–2 mm
+ LiDAR (FAST-LIO)	+210 mm/min	1–2 mm
+ AprilTags, head cam	+4 mm/min	< 0.3 mm

State estimation over a 2 min active stand (foot-anchoring residual): fiducials cut drift $\sim 60\times$.

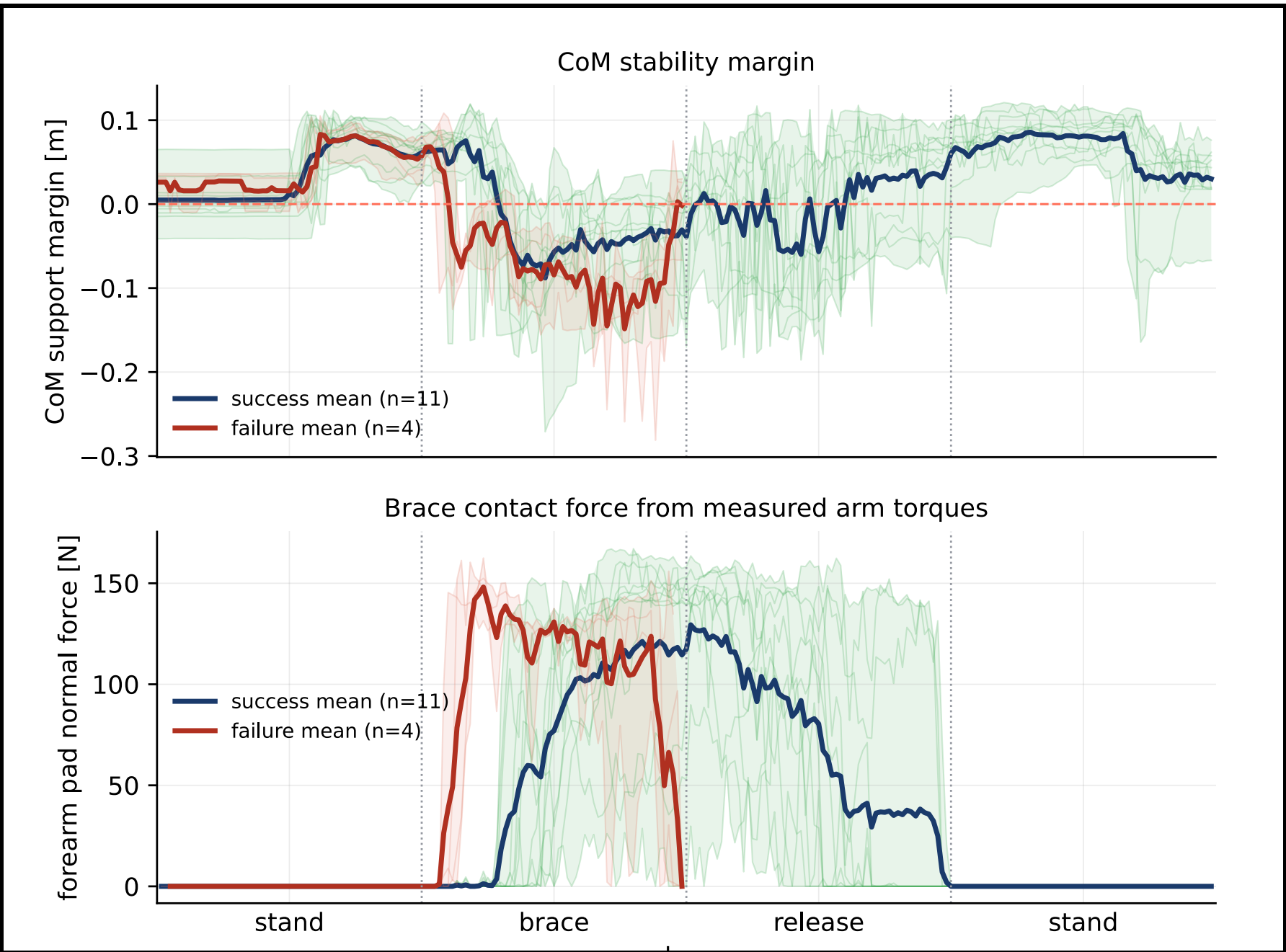
Brace cost ablation, stability, failure, and planner sensitivity



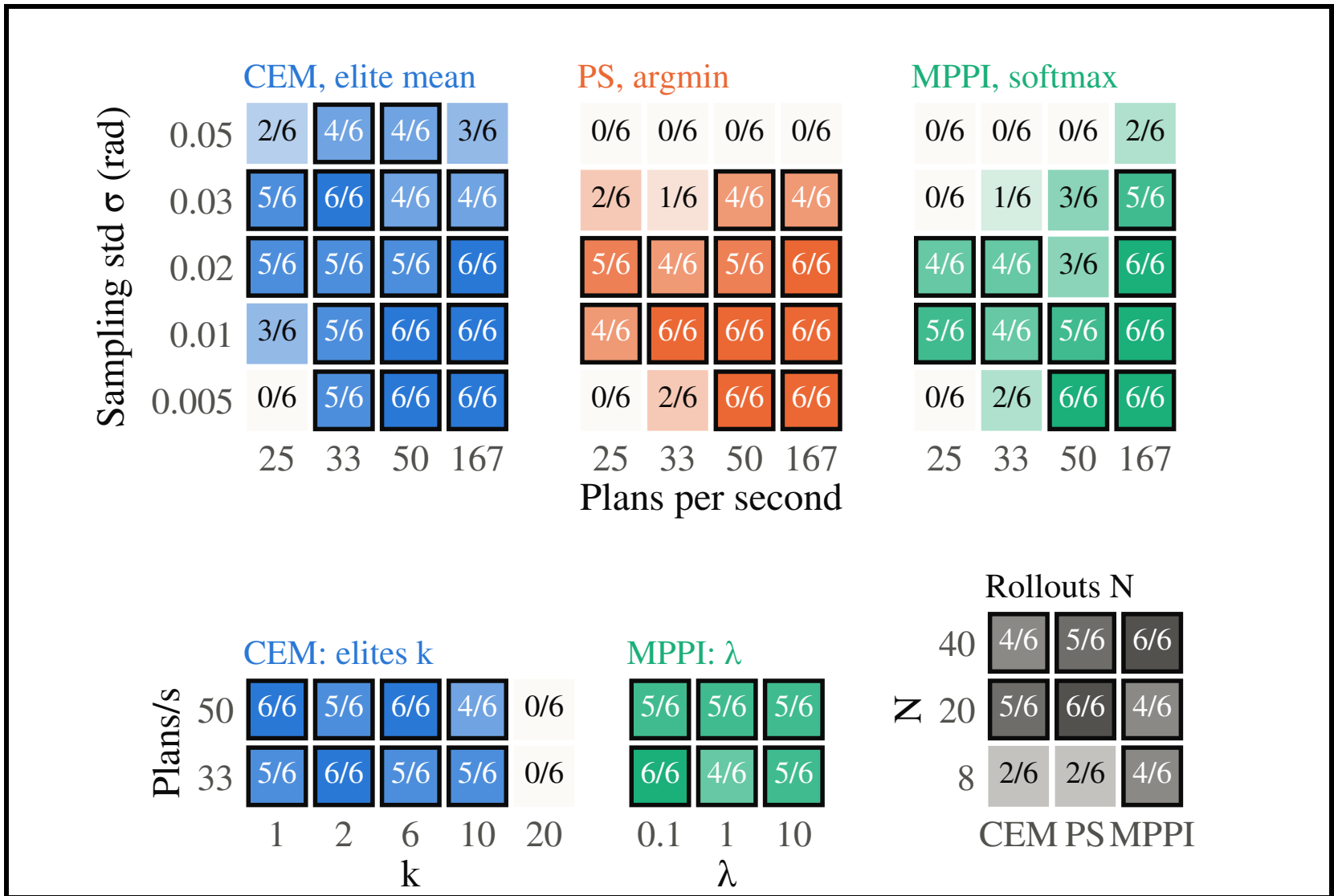
Brace-hold success on hardware vs. number of active brace cost terms.



Sim (orange) vs. hardware (green): reach and support margin agree; the real forearm carries less load (121 vs. 182 N).



15 max-reach trials, 11 successes. Successes load the brace progressively; failures load 150 N at onset and creep forward.



Completion over sampling noise and plan rate: CEM 16/20 cells, PS 12/20, MPPI 10/20; all fail at 10 Hz.